Response to the Reviewers

ORACLE: A Persistent Chemical State for Reproducible Molecular Workflows

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Dear Professor Zheng,

We thank you and the reviewers for the careful and constructive assessment of our manuscript. We have substantially revised the manuscript and added or clarified the following points. All changes are shown in red in the separate marked manuscript submitted for review. The unmarked manuscript is supplied as a separate clean file.

The central contribution is now stated directly: Operator for Recognition, Atom-typing and Continuous Local Equivalence (ORACLE) couples atomic synthons and the effective atomic number to a persistent, auditable chemical state that carries continuous geometric and electronic evidence into discrete, reusable workflow objects. This state is exercised across clustering, transition-state chart selection, open-shell and excited-state analysis, solvent-aware updates, overlapping-fragment assembly, and high-coordination chemistry. The principal revisions are: (i) a substantially reframed scientific claim presenting ORACLE as a persistent, auditable state-transfer layer for general chemical workflows; (ii) a narrower statement of the demonstrated domain; (iii) precise definitions of immutable snapshots, versioning, persistence, and invalidation; (iv) corrected level nomenclature and the saccharin atom-type description; (v) explicit separation of response uncertainty from within-class chemical dispersion; (vi) a quantitative held-out end-to-end structural-transfer example; (vii) a clearer comparison with existing cheminformatics tools and Proxima; and (viii) an expanded description of the ORACLE coordinate workflow and transition-state perception, and the implications of symmetry projection for downstream properties. We have also added a compact terminology table, reduced the main-text validation to four discriminating tests, moved the detailed benchmark tables to the Supporting Information, and made SWITCH explicit as the production chemoinformatics layer that replaces RDKit in the workflow. The seven-target leave-one-identity-out selector is now reported as a set, with thiazole shown as the representative case. The code-availability statement identifies the public versioned repository and the archived source. We have additionally added a non-obvious polynuclear coordination example, neutral singlet $Fe_4S_4Cl_4$ in C_{3v} symmetry, with its explicit 22-edge graph and independent rank/closure diagnostics. To avoid implying chemical significance beyond the underlying calculations, we have rounded numerical results in the main text to the precision supported by the corresponding validation, while retaining higher-precision values in the Supporting Information only where they are needed for auditability or exact reproduction. We have also documented the new explicit transition-state perception route, including condition-number selection among admissible reactive primitive charts. The validation now also spans state changes beyond the equilibrium closed-shell case: CH_2 singlet/triplet and O_2 triplet states, TDA/PBE0 S_1 densities for formaldehyde and uracil, and uracil S_0 with aqueous C-PCM. In uracil, the two carbonyl Mayer orders change from 2.072/2.037 in the gas phase to 2.049/1.977 with C-PCM, and to 1.966/1.828 in S_1. These results make state- and environment-dependent atom typing and force-field parameter selection explicit, extending the workflow beyond geometry-only perception. We also added a secondary efficiency check on eight systems spanning 3–176 atoms. The persistent ORACLE state and a Cartesian-input route that forces reperception use identical compiler settings and correctness checks. The persistent route is faster in all eight cases, with a median case-wise speed-up of 1.33 (the median elapsed times are 0.257 and 0.327 s), while the accepted chemical representation and downstream results are identical in every case. The complete case-level table is now in the Supporting Information. We also make the capability boundary with RDKit explicit. RDKit is an appropriate reference for equilibrium molecular graphs, whereas ORACLE retains forming and breaking contacts as typed transition-state evidence and selects among admissible primitive charts using the Wilson-B condition number. The transition-state examples demonstrate this capability and its transfer to downstream coordinate construction.

The revision also makes the central strengths and novelties explicit. The atomic synthon and the effective atomic number are ORACLE's central chemical module: the synthon is the local, continuous chemical state, while $Z_i^{\mathrm{eff}} = Z_i - 1/2 + \mathrm{ET}_i$ is its compact elemental–environment coordinate. Together they make chemical environments comparable, element-preserving, clusterable, and transferable across geometries and electronic states. A single declared QM closure at fixed geometry supplies CM5 charges, total and spin-resolved Mayer orders, and the electronic feedback needed by these descriptors; for larger systems the closure can be assembled from overlapping fragments, without an optimization step. The same state then supports synthon clustering, aromaticity, interactions, fragmentation, symmetry, transition-state charts, and coordination environments beyond ordinary valence, including the non-obvious 22-edge $Fe_4S_4Cl_4$ cluster. Persistence, dependency tracking, and provenance make these objects reusable by internal-coordinate optimization and by open-shell, excited-state, solvent, and TS workflows. The contribution is the general and executable coupling of this chemical module across workflow regimes, complementing rather than replacing individual electronic-structure or cheminformatics algorithms.

The revised manuscript now includes a compact capability table, a quantitative state-consequence table, and a central chemical-module schematic. A dedicated subsection, "Chemical consequences of continuous perception," makes explicit how atomic synthons and Z_i^{eff} support clustering, parameter transfer, reactive charts, electronic-state comparisons, solvent-aware analysis, and high-coordination chemistry. For a more immediate overview, we also added a non-TikZ editorial figure at the beginning of the manuscript. It places quantum-mechanical evidence, atomic synthons, Z_i^{eff}, state-specific analysis, transition-state charts, overlapping fragments, and high-coordination chemistry in one visual workflow. The text now states the four corresponding layers explicitly: QM evidence, persistent state, accepted discrete objects, and downstream operations. The conclusion was shortened accordingly so that the central message is stated once and the application range is connected directly to medicinal chemistry, photovoltaics, catalysis, and related molecular-materials workflows.

Response to the Editor

We have followed the request for two manuscript versions. The clean version contains no colored or highlighted text, while the marked version identifies the changes in red. Funding, ORCID, ACS formatting, references, and the author-created Table of Contents graphic should be checked again in the ACS submission form at upload; the revised manuscript retains the validated ORCID identifier and the original author-created graphic.

Response to Reviewer 1

General assessment. We appreciate the recognition that the implementation is technically competent and that the validation is extensive. The revision now makes clear why the range of applications is scientifically useful: continuous perception preserves graded geometric and electronic evidence, while discrete acceptance makes the resulting graphs, clusters, interactions, symmetry operations, reactive charts, and high-coordination objects executable and auditable. A single fixed-geometry QM closure, also available through overlapping fragments, supplies the CM5/Mayer/spin feedback that geometry-only perception cannot provide. The examples span equilibrium, open-shell, excited-state, solvent-dependent, transition-state, fragment, and polynuclear coordination regimes within this one contract. The intended role is to support sophisticated QM calculations for chemically important molecules in the roughly 100–200 atom range, including medicinal-chemistry, photovoltaic, catalytic and related molecular-materials workflows. The manuscript now connects this scale directly to the paclitaxel, vancomycin, [6]helicene and 176-atom state-transfer benchmarks already included in the validation.

Reviewer 1 assessment (verbatim).

> This manuscript presents ORACLE, a software framework for persistent molecular per-

ception designed to maintain a consistent chemical interpretation across computational workflows. The implementation appears technically competent, and the validation is extensive.

However, I have concerns about the practical significance, methodological necessity, and validation of the central concept of persistent molecular perception. The manuscript presents a software-infrastructure problem in highly abstract language, while providing limited evidence that such inconsistencies lead to scientifically consequential errors in realistic computational workflows.

The software appears potentially useful, but the authors need to provide stronger evidence that persistent molecular perception addresses a practically important problem, clarify its scientific contribution beyond software infrastructure, and better justify the L0 methodology and validation strategy. I therefore recommend major revision.

The revision addresses this assessment through the point-by-point responses below.

1. The manuscript argues that different workflow components may generate inconsistent molecular representations, but does not demonstrate how frequently this occurs or whether it leads to scientifically meaningful errors. The author should provide at least one quantitative, end-to-end example showing that inconsistent perception changes a downstream computational result. The relationship to existing cheminformatics toolkits such as RDKit, Open Babel, and CDK should also be discussed more explicitly.

Comment 1: Frequency and consequences of inconsistent perception

Response. We added two complementary quantitative thiazole demonstrations in the Results and Discussion. Test case V is a predeclared seven-target, leave-one-identity-out donor-retrieval test; for the representative thiazole, the bond RMS error decreases from 14.1 to 3.2 mÅ and the aligned Cartesian RMS error from 14.5 to 5.1 mÅ. Test case VI then exercises the complete SMILES-to-structure route: initial geometry generation and GFN2-xTB optimization, persistent ORACLE perception for LCB25 fragment retrieval, differential correction, and final re-perception. In that distinct protocol, the bond RMS error decreases from 13.3 to 4.9 mÅ and the aligned Cartesian RMS error from 16.1 to 8.6 mÅ. The Introduction summarizes the first, leave-one-identity-out demonstration, while the Results and Discussion present both cases explicitly. Together they show measurable downstream structural consequences of preserving topology, aromatic assignment, and primitive-coordinate identity across workflow stages.

We also expanded the comparison with RDKit, Open Babel, and CDK. These toolkits provide important application-level operations and are treated as independent references or interoperable services. ORACLE's distinct claim is the shared, versioned state and its dependency-aware handoff to downstream calculations, not ownership of every cheminformatics operation.

2. It is not clear which advantages arise specifically from persistent molecular states rather than from conventional molecular data structures, provenance tracking, and workflow management. The authors should demonstrate what chemically consequential problems ORACLE solves that existing approaches cannot readily address.

Comment 2: Advantages beyond conventional data structures and provenance

Response. We clarified that provenance alone is not the proposed advance. The chemical state couples continuous evidence (coordination, charge, bond order, and local descriptors) to explicitly accepted discrete objects (graph, rings, symmetry actions, fragments, and interactions). Each object records its dependencies and is either reused or selectively invalidated when a declared input changes. This prevents a downstream application from silently rebuilding a different graph or symmetry action.

The revised manuscript now gives concrete chemically consequential examples: the same state is

used for aromaticity and symmetry, hydrogen-bond interaction objects remain separate from covalent identity, overlapping fragments preserve charge and atom mapping, and structural correction is followed by audited re-perception. These examples complement the component-level conformance audits with a workflow-level demonstration of persistent state transfer.

> **3.** The manuscript uses PBE0/def2-TZVP to generate CM5 charges and Mayer bond orders, but does not demonstrate that this step materially improves molecular perception. A comparison of geometry-only perception with and without L0 feedback would be valuable. If L0 is mainly a fallback for generating missing electronic descriptors, the author should state this explicitly rather than presenting it as an essential component of perception.

Comment 3: Whether L0 materially improves perception

Response. We agree that the role of L0 should not be overstated. The revised text now states explicitly that L0 is an optional electronic closure and feedback route. Geometry-only perception remains available when a complete state is supplied or when topology, rings, symmetry, and geometric descriptors are requested. L0 is needed in the reported examples only to supply missing CM5, Mayer, and spin-resolved observables and to demonstrate the reverse perception–QM feedback loop. Agreement with RDKit is therefore not presented as evidence that L0 is essential.

> **4.** The author describes PBE0/def2-TZVP level as a consistent baseline, but no sensitivity analysis is provided. It would be useful to show whether the relevant perception results are robust with respect to the functional and basis set, and whether a substantially cheaper level could provide comparable information. The computational cost of the L0 procedure should also be quantified, particularly for larger systems.

Comment 4: Functional, basis-set, and computational-cost sensitivity

Response. We narrowed the wording around PBE0/def2-TZVP: it is a reproducible operational baseline for the reported electronic closure, not a universally optimal level. We also distinguish the optional L0 single-point step from geometry-only perception and state that a cheaper level may be appropriate for applications that do not require the reported electronic observables.

The revised manuscript reports the measured same-host perception timings separately from the electronic calculation and adds the three-fragment Gly–Ala–Gly overlap reconstruction as a concrete large-state electronic- closure example. The three fragments cover all 27 atoms and 26 bonds with one fixed-geometry QM calculation per fragment, giving a CM5 RMS error of 0.0032 e and a maximum error of 0.0101 e relative to the full-molecule calculation. Together with the 42-atom five-tile [6]helicene reconstruction, this establishes overlap-fragment assembly as an operational route for larger systems while preserving ownership, boundary diagnostics and QM metadata.

> **5.** The validation corpus is extensive, but much of it tests individual perception components rather than persistence across workflows. For example, agreement with RDKit demonstrates consistency for aromaticity perception, but does not demonstrate that ORACLE maintains a chemically meaningful state as a molecule passes through different computational procedures. The authors should add at least one or two explicit cross-workflow or trajectory-based demonstrations.

Comment 5: Cross-workflow and trajectory-based validation

Response. We added explicit cross-workflow language and examples. ORACLE owns topology, rings, symmetry, and the primitive-coordinate source; coordinate construction performs internal-to-Cartesian realization and optimization; the resulting geometry returns to ORACLE for dependency-aware reanalysis. Downstream sampling selects points in active-variable space without reconstructing the chemical state.

The revised validation also includes the held-out thiazole structural-transfer test described above, overlapping electronic-fragment reconstruction for Gly–Ala–Gly, explicit large-ring and symmetry audits, and a neutral $Fe_4S_4Cl_4$ C_{3v} coordination case in which the enriched record preserves a non-obvious 22-edge polynuclear graph and reaches rank 30/30. These are cross-workflow demonstrations that complement the component-level audits and test persistence across geometry, electronic-structure and coordination operations.

We have also added the newly implemented transition-state capability. For an explicit single-geometry TS request, ORACLE retains possible breaking and forming contacts as typed evidence without changing the constitutional graph, constructs the reactive-zone candidate charts, and compares admissible primitive combinations using the normalized Wilson-B condition number. When the declared trigger and material-gain criteria are met, the combination with the lowest condition number is selected and the decision is recorded in the TS provenance. The coordinate layer then consumes this ORACLE-owned prescription. The revised manuscript now makes the transition-state capability explicit. The 25 geometries span ring opening, dissociation, insertion, torsional isomerization, and rearrangement, demonstrating that the feature is not tied to a single transition-state topology.

> **6.** The distinction between ORACLE and the author's previous Proxima framework is not sufficiently clear. A concise comparison of the capabilities and genuinely new methodological elements would help establish the novelty of the present work.

Comment 6: Distinction from Proxima

Response. We added a direct comparison in the Introduction and Computational Details. Proxima supplied geometry-first continuous covalent and hydrogen-bond descriptors coupled to ring analysis. ORACLE retains that continuous perception lineage but adds accepted versioned chemical objects, explicit dependency invalidation, provenance-bearing symmetry and permutation actions, fragment views, electronic feedback, and the ORACLE coordinate-transport contract. The revised text presents these as the genuinely new integrated methodological elements.

> **7.** The discussion of symmetry projection should address whether coordinate changes introduced by projection can affect downstream properties such as frequencies or reaction barriers.

Comment 7: Effect of symmetry projection on frequencies and barriers

Response. We added an explicit caveat. Symmetry projection is a coordinate transformation whose maximum displacement is recorded; it is not assumed to leave all downstream properties unchanged. In the quasi-O_h cubane audit, the maximum displacement is 3.3×10^{-4} Å and strict reanalysis restores all 48 operations with a residual below 10^{-15} Å. Frequencies, reaction barriers, and other derivative properties must be recomputed on the projected geometry, or the projection must be omitted when the displacement is chemically relevant.

Response to Reviewer 2

General assessment. We thank the reviewer for identifying places where the language needed tighter definition. We have clarified that the paper is a computational-chemistry workflow methodology centered on a chemical representation module. Atomic synthons and the effective atomic number convert continuous geometric and electronic evidence into comparable local identities; persistence, discrete acceptance, and provenance make those identities executable and auditable. The same module spans synthon clustering, transition-state chart selection, open-shell and excited-state feedback, solvent-aware state updates, fragment assembly, and non-obvious high-coordination topology, including $Fe_4S_4Cl_4$. The statistical quantities and the validation domains are stated explicitly.

Reviewer 2 assessment (verbatim).

> The manuscript presents ORACLE, a computational framework designed to maintain a continuous, persistent, and auditable chemical state connecting molecular perception with quantum-chemical calculations and downstream applications. The framework integrates topology, aromaticity, atom typing, symmetry, fragmentation, electronic descriptors, and structural transfer within a common versioned representation, and is validated using aromatic systems, hydrogen-bonded dimers, open-shell species, fragment assemblies, and structural reconstruction tests.
>
> The concept is ambitious and potentially useful, but the scientific claims are broader than the presented validation supports. Moreover, the principal contribution appears closer to cheminformatics, molecular representation, and computational workflow architecture than to a new theory or electronic- structure methodology. In its present form, the work may therefore be more naturally suited to a journal such as Journal of Chemical Information and Modeling. If the manuscript is to be considered further in JCTC, substantial additional methodological validation would be required.

Scope and relationship to cheminformatics. Response. We clarified that ORACLE is a computational chemistry method built around a molecular representation contract, rather than a new electronic structure theory. Its central chemical module is the atomic synthon together with the effective atomic number $Z_i^{\mathrm{eff}} = Z_i - 1/2 + \mathrm{ET}_i$: continuous geometry, coordination, charge, bond-order, and spin evidence is organized into local, element-preserving chemical identities that can be compared, clustered, and transferred. Discrete acceptance, persistence, dependency tracking, and provenance make those identities executable across graphs, rings, interactions, symmetry operations, reactive charts, and high-coordination objects. A single fixed-geometry QM closure supplies CM5, Mayer, and spin observables, and the same closure can be assembled from overlapping fragments for larger systems. This common state supports the ORACLE coordinate workflow, including the 25 transition-state cases, and the non-obvious 22-edge $Fe_4S_4Cl_4$ coordination example. We now describe the relationship to RDKit, Open Babel, and CDK more explicitly and position ORACLE as complementary to their individual graph and cheminformatics operations. The revised Introduction and the new capability table now make the scope-fit concrete: the QM closure supplies electronic observables that change atom typing, parameter transfer and reactive-coordinate selection, and the same state is exercised on open-shell, excited-state, solvent, TS and high-coordination cases. This is the direct JCTC rationale: the state is the computational substrate for efficient metrics, chemistry-guided overlapping fragmentation, accurate force-field parameter transfer and QM-informed modelling at molecular scale. The effective atomic number is in the conceptual lineage of atoms-in-molecules descriptions, but couples the local atomic object to continuous geometry, coordination, charge, bond-order and spin evidence and makes it executable across a complete workflow.

Concretely, this is embodied in three methodological elements already present in the manuscript: (i) the 4×5 hydrogen-bond response table (Section 3.4, Figure 4), a directly deployable QM-derived parameter set in the tradition of RESP-type charge-derivation methodology; (ii) the differential structural-correction formalism (Eqs. 18–19), which provides a quantitative QM-to-geometry transfer route; and (iii) the Wilson-B condition-number selection rule for transition-state charts (Section 3.5), which is geometry- optimization methodology rather than descriptive cheminformatics. Together with the open-shell, excited-state, solvent and $Fe_4S_4Cl_4$ tests, these elements make the JCTC scope directly verifiable from the manuscript.

> **1.** The central claim is not directly tested. The manuscript argues that the combination of continuous perception, discrete acceptance, persistence, provenance, and dependency tracking is the main advance, but no ablation study or controlled comparison demonstrates that this combination improves a scientific result compared with a simpler conventional workflow.

Comment 1: Central claim and ablation

Response. We now separate the claims tested by the ablation. Eight systems spanning 3–176 atoms were passed through the same compiler either from a validated ORACLE state or from identical Cartesian input that forced reperception. The persistent route was faster in all eight cases (median 0.257 versus 0.327 s; median case-wise speed-up 1.33-fold), with identical accepted chemical representations and downstream results. This result establishes computational reuse, handoff consistency and reproducibility for the deterministic workflow. The held-out thiazole workflow transfers accepted objects through generation, retrieval, correction and re-perception, while the transition-state corpus exercises reactive charts that a standard equilibrium-graph route cannot represent. Together, these experiments show how persistent state preserves reproducible chemical objects while continuous electronic and reactive evidence changes the downstream interpretation when the state changes. The complete ablation data are reported in the Supporting Information JSON record.

Response. The ablation targets computational reuse and reproducibility specifically (the Efficiency row in Table 5). It is not intended to test whether ORACLE succeeds where conventional heterogeneous pipelines cannot execute the task. That capability claim is supported separately by the transition-state and $Fe_4S_4Cl_4$ tests, while the state-blind electronic control measures the consequence of retaining electronic state at fixed geometry.

> **2.** The concepts "persistent", "immutable", and "perceived once" are used inconsistently. Elsewhere, the state is explicitly updated, the refined geometry is "perceived again", and assembled or optimized structures are re-perceived. The authors should define precisely what is immutable, what is versioned, and under what conditions chemical identity is considered unchanged.

Comment 2: Persistent, immutable, and perceived-once state

Response. We now define an immutable state as an immutable serialized snapshot, not an object that can never be updated. A change in a declared dependency creates a new version with a parent fingerprint; unchanged dependencies permit reuse. Accordingly, "perceived once" means once per state version, while "persistent" means persisted and reused until targeted invalidation.

> **3.** The level notation is internally inconsistent. Section 3.5 defines only L0, L1, and PL1 and gives the sequence L0 $\rightarrow$ L1 $\rightarrow$ PL1, whereas Figure 10 uses L1, BL1, and PL2. BL1 and PL2 are not consistently defined in the main text.

Comment 3: Inconsistent level notation

Response. We unified the notation throughout the manuscript and Supporting Information as L0, L1, and PL1 for the operational route. The identifiers BL1 and PL2 have been removed from the manuscript, Supporting Information, distributed file names and exposed workflow descriptions. The corresponding geometries are now described functionally as a frozen L1 structure and a higher-level reference structure, without introducing additional perception levels.

> **4.** The saccharin GAFF2 discussion contains an apparent chemical inconsistency. The text refers to a "hydroxyl oxygen" mapped as "oh", whereas Figure 10 shows an "o5" sulfur-bound oxygen, and saccharin does not contain the described hydroxyl group. This should be carefully checked in a paper focused on reliable atom perception and typing.

Comment 4: Saccharin atom typing

Response. Saccharin's relevant oxygen in the cited mapping is sulfur-bound. The mapping is now stated as aromatic carbon, carbonyl carbon, sulfur-bound oxygen, amide nitrogen, and hypervalent sulfur mapped to 'ca', 'c', 'o5', 'n', and 's6', respectively; the carbonyl oxygen is mapped to the distinct 'o' class.

5. Several statistical and validation claims appear overstated. For the hydrogen-bond model, the abstract highlights uncertainties of up to 0.0085 e, whereas the main text reports within-class standard deviations of up to 0.0168 e for charge transfer. These quantities measure different aspects of variability, and the current wording may understate the underlying chemical dispersion. Likewise, 15 doublets across seven classes do not appear sufficient to support the claim of "statistically resolved coverage" of a broad open-shell domain. These claims should be narrowed and the statistical measures more clearly distinguished.

Comment 5: Statistical claims and open-shell coverage

Response. We now distinguish the hydrogen-bond response uncertainty from the within-class chemical dispersion. Section 3.4 of the manuscript reports a leave-one-environment-out response uncertainty up to 0.0085 e, while the within-class charge-transfer standard deviation reaches 0.0168 e. These are different quantities and are no longer conflated. The open-shell corpus is described as a stratified set of 18 neutral doublets in nine declared electronic classes, with two observations per class. The corpus-level intervals quantify the range and reproducibility of the electronic closure; the two observations per class provide a descriptive within-class comparison, not an independent estimate of class-level uncertainty, and establish a state-resolved reference for the downstream synthon and effective-atomic-number updates.

We have also added state-specific validation. On the same ORACLE-perceived geometry, CH_2 is compared in singlet and triplet states, O_2 is treated as a triplet, and formaldehyde and uracil are evaluated in their singlet ground states and with TDA/PBE0 S_1 densities. Uracil is additionally compared in the gas phase and with aqueous C-PCM. The resulting CM5-charge and Mayer-order changes are reported in the manuscript and Supporting Information, including the uracil carbonyl changes 2.072/2.037 to 1.966/1.828 in S_1 and to 2.049/1.977 with C-PCM. These are direct same-geometry comparisons: a graph-only perception layer would retain the same nuclear connectivity, while ORACLE retains multiplicity, electronic-state and dielectric-environment context and propagates the changed CM5, Mayer and spin observables into the chemical state. The same principle is essential for non-obvious open-shell species, for which connectivity alone does not identify the chemically relevant state.

In addition, we performed a functional/basis sensitivity calculation with PBE0/def2-SVP, PBE0/def2-TZVP, and B3LYP/def2- TZVP on the same ORACLE geometries. For formaldehyde, the S_1 C=O Mayer orders are 1.459, 1.445, and 1.430, compared with S_0 values of 2.106, 2.098, and 2.092; for uracil, the PBE0/def2-TZVP S_1 carbonyl values are 1.966/1.828 versus 2.072/2.037 in S_0. The complete sensitivity table and excitation energies are now in the Supporting Information. These results quantify the operational L0 range and support retaining functional, basis, state, and solvent metadata rather than presenting PBE0/def2-TZVP as an uncertainty-free reference.

The new calculations quantify the operational sensitivity of the L0 closure and show why functional, basis, electronic-state, and solvent metadata belong to the persistent chemical state. They also show that the state effect is larger than the functional/basis variation for the reported formaldehyde and uracil comparisons, making the preserved metadata chemically actionable.

Closing

The revised manuscript now presents ORACLE as a chemical-state methodology for integrating sophisticated QM calculations with reproducible molecular workflows. Its central objects—atomic synthons and the effective atomic number—connect continuous QM evidence to auditable discrete operations, including clustering, force-field parameter transfer, fragmentation, transition-state charts, excited-state and solvent-aware analysis, and high-coordination chemistry. The new figure and the streamlined conclusion make this scope explicit while the point-by-point responses document the

corresponding evidence and revisions. We thank the reviewers for helping us improve the scientific positioning and technical clarity of the manuscript.

Sincerely,

Vincenzo Barone